

AYUSH RANJANE

AI/ML ENGINEER | B.E. Artificial Intelligence & Data Science (2027)

Pune, Maharashtra, India | +91-9322623848 | ayushranjane@gmail.com

linkedin.com/in/ayush-ranjane-61051b303 | github.com/Ayush-Ranjane | ayushranjane.in



PROFESSIONAL SUMMARY

Engineering practical AI solutions as a B.E. Artificial Intelligence & Data Science student transitioning into a Graduate Engineer Trainee role. Developing **end-to-end machine learning applications** using a robust stack of Python, SQL, Scikit-learn, and PostgreSQL. Deploying data-driven systems with hands-on expertise in **predictive modeling, anomaly detection, feature engineering, and REST API development**.

TECHNICAL SKILLS

Programming	Python, SQL
Machine Learning	Scikit-learn, XGBoost, Time-Series Forecasting, Anomaly Detection, Feature Engineering, Model Evaluation
Data Science	Pandas, NumPy, Exploratory Data Analysis, Statistical Analysis, Data Visualization, ETL
Backend / APIs	Flask, FastAPI, REST APIs
Databases / Tools	PostgreSQL, MongoDB, Git, GitHub, PowerBI
Frontend	JavaScript, HTML, CSS

PROJECTS

GreenCO₂ — AI-Powered CO₂ Monitoring & Compliance Platform

Jan 2026 – Mar 2026

Python, Flask, PostgreSQL, React, Prophet, Isolation Forest, Docker, JWT, REST APIs | Live Demo

- Engineered an end-to-end CO₂ monitoring and forecasting pipeline using **Python, PostgreSQL, Prophet, and Scikit-learn**, achieving **94% forecasting accuracy** for 7–14 day emission predictions.
- Developed an **Isolation Forest anomaly detection pipeline** to identify abnormal emission patterns across historical multi-source data using source-level emissions, temporal trends, and threshold-based features.
- Automated **data preprocessing, feature engineering, model training, evaluation, and prediction workflows**, enabling company-specific models and continuous ML-driven emission monitoring.

AQI Prediction System — Air Quality Index Forecasting Pipeline

Jan 2026

Python, XGBoost, Random Forest, Scikit-learn, Flask, Pandas, NumPy | Live Demo

- Developed an end-to-end AQI prediction pipeline processing **11 pollutant parameters**, including PM2.5, PM10, NOx, CO, and O₃, with preprocessing, EDA, feature engineering, and feature selection.
- Trained and evaluated **XGBoost, Random Forest, and Linear Regression** models, achieving an **R² score of 0.92** and **RMSE below 2.5**.
- Deployed the selected model through a **Flask-based web interface** that transforms pollutant inputs into interpretable AQI risk categories in real time.

EDUCATION

B.E. — Artificial Intelligence & Data Science

2023 – 2027

Dr. D. Y. Patil Institute of Engineering, Management & Research (DYPIEMR), Pune | CGPA: **8.53 / 10**

Higher Secondary (12th Standard) — Computer Science

2021 – 2023

Dnyansamwardhini Junior College, Shirwal, Satara | Bifocal Stream

Secondary (10th Standard)

2021

Dnyansamwardhini Vidyalaya, Shirwal, Satara

LEADERSHIP

Joint Technical Head — Novus Neuron Club, DYPIEMR

Sep 2025 – Sep 2026

- Guided junior members in **data preprocessing, machine learning model development, evaluation, and project troubleshooting**.
- Provided structured **technical debugging and implementation support** for AI/ML projects developed by club members.

CERTIFICATIONS

IBM Machine Learning Professional Certificate — IBM via Coursera

IBM Data Science Professional Certificate — IBM via Coursera

DSA for Deeptech — Brainovision Solutions India Pvt. Ltd. & AICTE